

Slope or y -Intercept? Find the Mix-Up

Directions

- In $y = mx + b$, the number *multiplying* the variable is the slope: how much y changes for each 1 unit of x . The number *added on* is the y -intercept: the value of y when x is 0.
- A quick test for the intercept: substitute $x = 0$ and see what is left.
- When you name an error, say which number belongs in which place and why — a corrected equation on its own is not an explanation.

1. A gym charges $C = 12x + 40$ dollars for x months of membership.

(a) What is the slope — the change in cost for each extra month?

Answer: _____

(b) What is the y -intercept — the cost before any months are used?

Answer: _____

2. A line passes through $(0, 4)$ and $(5, 19)$. Tomas wrote its equation as $y = 4x + 3$.

(a) Find the slope of the line.

Answer: _____

(b) Using the correct equation, find y when $x = 2$.

Answer: _____

(c) Name the error in the equation Tomas wrote.

Answer: _____

3. Nia says that the graph of $y = 7x + 2$ crosses the y -axis at 7 and rises 2 units for each 1 unit across.

(a) Find the value of y where the graph crosses the y -axis.

Answer: _____

(b) Find how much y rises for each 1 unit increase in x .

Answer: _____

(c) Name the error in the description Nia gave.

Answer: _____

4. A phone plan costs \$25 when 0 gigabytes are used and \$45 when 5 gigabytes are used. Reza wrote the cost equation as $C = 25g + 4$ dollars for g gigabytes.

(a) Find the cost per gigabyte.

Answer: _____

(b) Using the correct equation, find the cost when 8 gigabytes are used.

Answer: _____

(c) Name the error in the equation Reza wrote.

Answer: _____

- 5.** A taxi's total charge is shown in the table. Kai says the equation is $y = 3x + 4$ dollars for x miles.

Miles	0	2	4	6
Total charge (\$)	3	11	19	27

- (a) What is the charge at 0 miles?

Answer: _____

- (b) Find the charge per mile.

Answer: _____

- (c) Name the error in the equation Kai wrote.

Answer: _____

- 6.** A candle is 20 cm tall and burns down 3 cm per hour, so its height after t hours is $h = 20 - 3t$ centimetres. Lena wrote $h = 3 - 20t$ instead.

- (a) Using the correct equation, find the candle's height after 4 hours.

Answer: _____

- (b) Name the error in the equation Lena wrote.

Answer: _____

7. Two lines are given by $y = 5x + 2$ and $y = 2x + 5$.

- (a) Compare their slopes using $>$ or $<$, first line then second.

Answer: _____

- (b) Compare their y -intercepts using $>$ or $<$, first line then second.

Answer: _____

- (c) Explain why the two equations describe different lines even though they are built from the same two numbers.

Answer: _____

8. A savings account holds \$60 at week 0 and \$132 at week 8. Marcus wrote the balance equation as $y = 60x + 9$ dollars after x weeks.

- (a) Find the amount saved per week.

Answer: _____

- (b) Using the correct equation, find the balance at week 10.

Answer: _____

- (c) Explain how you could tell that the balance Marcus would predict for week 10 is far too large, without doing his calculation.

Answer: _____